

Arun Vignesh Malarkkan

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🌐 <http://arunvignesh28.github.io/>

About Me

- 📌 Investigating reliability and generalization of LLM reasoning, combining causal inference and mechanistic interpretability, focusing on invariance discovery, spurious correlation mitigation, and robust evaluation.

Education

Aug 2023 – Present
(Expected: Spring 2027)

- 📌 **Ph.D., Computer Science**, Arizona State University, Tempe, AZ.
Advisor: Dr. Ross Maciejewski.
CGPA: 3.9/4.0.

May 2020

- 📌 **Master of Science, Computer Science**, Arizona State University, Tempe, AZ.
CGPA: 3.8/4.0.

Technical Skills

Core Expertise	📌 LLM Reasoning & Reliability, Recommendation Systems, Causal Inference & Causal Representation Learning, Mechanistic Interpretability, Post-training & Fine-tuning, Retrieval-Augmented Generation (RAG), Data-Centric AI.
ML & Systems	📌 PyTorch, JAX, TensorFlow, Hugging Face Transformers, Distributed Training, GPU/CUDA Programming.
LLM & Generative AI	📌 PEFT (LoRA, QLoRA), LangChain, LlamaIndex, OpenAI API, vLLM.
Interpretability & Analysis	📌 TransformerLens, Activation Patching, Probing, PyTorch Hooks.
Programming	📌 Python, Java, TypeScript.
Data & Infrastructure	📌 AWS, GCP, BigQuery, MySQL, Apache Spark (SparkML), Docker.
General Libraries	📌 Scikit-learn, Pandas, NumPy, Flask, AutoGluon.

Research Publications

First-Author Publications

- 1 **A. V. Malarkkan**, U. Byahut, M. R. Choudhury, Y. Charde, V. Gupta, and Y. Fu, “Invariant reasoning directions in latent trajectories of language models,” in *ICML Workshop on Mechanistic Interpretability*, 2026.
- 2 **A. V. Malarkkan**, X. Wang, K. Liu, D. Zhang, and Y. Fu, “Causally-guided diffusion for stable feature selection,” 2026, arXiv preprint. Under review, KDD.
- 3 **A. V. Malarkkan**, W. Ying, and Y. Fu, *CAFE: Causally-guided automated feature engineering with multi-agent reinforcement learning*, arXiv preprint. Under review, NeurIPS, 2026.
- 4 **A. V. Malarkkan** et al., *FinRule-Bench: A benchmark for joint reasoning of llms over financial tables and principles*, arXiv preprint. Under review, AACL, 2026.
- 5 **A. V. Malarkkan** et al., *Latent reasoning does not lack signal, it lacks invariance: Subspace-constrained refinement for robust latent reasoning*, arXiv preprint. Under review, NeurIPS, 2026.
- 6 **A. V. Malarkkan**, H. Bai, A. Kaushik, and Y. Fu, “DELTA: Variational disentangled learning for privacy-preserving data reprogramming,” in *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, 2025.
- 7 **A. V. Malarkkan**, H. Bai, D. Wang, and Y. Fu, “Causal graph profiling via structural divergence for robust anomaly detection in cyber-physical systems,” in *Proceedings of the IEEE International Conference on Big Data*, 2025.

- 8 **A. V. Malarkkan**, D. Wang, H. Bai, and Y. Fu, "Incremental causal graph learning for online cyberattack detection in cyber-physical infrastructures," *IEEE Transactions on Big Data*, 2025.
- 9 **A. V. Malarkkan**, D. Wang, and Y. Fu, "Multi-view causal graph fusion based anomaly detection in cyber-physical infrastructures," in *Proceedings of the ACM International Conference on Information and Knowledge Management (CIKM)*, 2024.

Co-Authored Publications

- 1 H. Cao et al., *Not all correct gradients should be trusted: Shortcut-aware reasoning via directional detection and correction*, arXiv preprint. Under review, NeurIPS, 2026.
- 2 H. Cao et al., *Robust data reshaping via flat and consensus-seeking diffusion*, arXiv preprint. Under review, NeurIPS, 2026.
- 3 X. Wang, K. Liu, **A. V. Malarkkan**, and Y. Fu, *Evolving demonstration optimization for chain-of-thought feature transformation*, arXiv preprint. Under review, KDD, 2026.
- 4 W. Ying et al., *Distribution shift aware neural tabular learning*, arXiv preprint. Under review, ICML, 2026.
- 5 W. Ying et al., *A survey on data-centric ai: Tabular learning from reinforcement learning and generative ai*, arXiv preprint., 2025.

Professional Experience

- May 2026 – July 2026  **Research Intern : LLM Post-Training**, The Home Depot.
- Post-training 4B and 8B small language models for product compatibility reasoning, enabling catalog-scale verification of whether products can be correctly used, installed, or purchased together.
 - Developing a novel preference optimization framework that isolates stable compatibility-relevant learning signal from unstable catalog-specific variation during post-training, improving robustness to noisy metadata, paraphrased queries, and product taxonomy drift.
 - Building trajectory- and prediction-level evaluation protocols that stress-test compatibility reasoning across semantically equivalent product-pair queries, measuring accuracy, invariance, calibration, and consistency under distribution shift.
- Jun 2025 – Present  **Doctoral Researcher**, DOW Chemicals (National Academy of Engineering Grant).
- Leading research on incorporating LLM-guided knowledge priors into neuro-symbolic, causally-aware multi-agent reinforcement learning frameworks to encode domain constraints and guide structured reasoning in materials discovery.
 - Developing frameworks that integrate causal structure with sequential decision-making to improve robustness under distribution shifts in industrial simulation environments.
 - Designing interpretable and scalable models for material optimization under uncertainty (details subjected to confidentiality).
- Jan 2023 – Jun 2023  **Senior Software Engineer**, Fidelity Investments.
- Led development of Monte Carlo-based financial projection systems for retirement planning at scale.
 - Built probabilistic simulation frameworks for uncertainty-aware decision modeling in long-term financial forecasting.
 - Developed ensemble-based outlier detection models to identify key drivers affecting projected savings outcomes.

Professional Experience (continued)

- Jun 2021 – Jan 2023 **Software Development Engineer**, Amazon — Alexa AI.
- Developed and deployed purchase likelihood modeling for Alexa recommendation systems, improving voice-driven purchases by +8%.
 - Built end-to-end decision and ranking pipelines for voice-based commerce, integrating user behavior modeling with real-time inference systems.
 - Designed production systems for voice-enabled transactional flows with compliance constraints (e.g., parental consent, HIPAA-aligned systems).
- Jun 2020 – May 2021 **Data Scientist / Software Developer**, ASU Decision Center for Educational Excellence.
- Developed predictive models for demographic, financial, and infrastructure planning using large-scale public datasets.
 - Designed end-to-end data pipelines (AWS, ETL) for data ingestion, modeling, and deployment.
 - Built models for resource allocation and policy analysis, identifying key drivers of graduation rates and community-level outcomes.

Academic Research Experience

- 2024 – Present **Mechanistic Interpretability of LLM Reasoning: Causal Mediation and Latent Interventions**
- Developed a causal mediation framework over intermediate reasoning tokens using activation patching and counterfactual inputs; showed that a sparse subset of tokens acts as stable causal mediators of the final answer, yielding an interventional account of where reasoning becomes causally committed inside the forward pass.
 - Designed a training-free latent intervention method that identifies invariant reasoning directions via SVD of contrastive latent trajectories and constrains refinement to this subspace with an adaptive alignment gate; outperforms all latent refinement baselines on six reasoning benchmarks (+2.1% average), improves paraphrase consistency by ~10%, and reduces latent trajectory variance by up to 50%, with zero parameter updates.
- 2025 – Present **Invariance-Constrained LLM Post-Training and Robust Evaluation**
- Developed a subspace-constrained preference optimization method that decomposes DPO updates into invariant task-relevant directions and unstable dataset-specific variation; deployed in production for catalog-scale product compatibility reasoning, improving incompatibility recall under paraphrase and distribution drift by 8% over standard DPO.
 - Built causal-counterfactual evaluation protocols and trajectory-level stress tests that separate rule-following from pattern matching, measuring accuracy, calibration, and consistency across semantically equivalent inputs under perturbation and taxonomy drift.
- 2023 – 2025 **Causal Foundations: Invariant Structure Learning under Distribution Shift**
- Established the methodological basis for the agenda above: incremental and multi-view causal graph learning for non-stationary systems (IEEE TBD, CIKM, IEEE BigData), plus causally guided feature construction and privacy-aware transformation achieving ~9% performance gain with ~35% privacy leakage reduction.

Teaching, Mentorship & Academic Service

Invited Talks

Dec 2025

■ **Keynote** — IEEE International Conference on Modelling, Simulation & Intelligent Computing (MoSICom), BITS Pilani, UAE.

"Beyond Correlation: Can Causal Reasoning Enhance Modern AI Models in Robustness and Interpretability?"

Presented research on causal inference, robustness, and interpretability in modern AI systems, emphasizing limitations of correlation-based learning and the role of causal reasoning.

Teaching

Aug 2023 – Present

■ **Senior Teaching Assistant / Lab Instructor**, Arizona State University.

- Led instruction and lab sessions for courses in Object-Oriented Programming, Data Mining, and Semantic Web Mining for large undergraduate cohorts, and also project management for 30 Capstone Project teams.
- Designed and developed hands-on lab modules and course materials, emphasizing applied machine learning and data-driven reasoning.

Mentorship

Jan 2024 – Present

■ **Research Mentor**, Arizona State University.

- Mentoring undergraduate and graduate researchers on Neuro-symbolic AI for feature engineering, world model simulations, and LLM reasoning optimization.
- Guiding projects on investigating efficient fine-tuning and evaluation of large language models for reasoning tasks, including experimental design and model analysis.

Academic Service

Jan 2025 – Present

■ **Program Committee Member:** AAAI 2026; IEEE Big Data 2026; IEEE ICSCAN 2024; IEEE BigData 2024 (Undergraduate & High School Research Symposium); IEEE ICDM 2025.

Conference Reviewer: KDD, ICLR, NeurIPS, ICDM, AAAI, IEEE BigData, Urban AI.

Journal Reviewer: ACM Transactions on Knowledge Discovery from Data (TKDD).

Session Chair: IEEE MoSICom 2025; IEEE ICDM 2025 — Undergraduate & High School Research Symposium.

Workshops & Outreach

■ **Outreach**

- Conducted hands-on workshops on Machine Learning fundamentals and Graph Neural Networks, training undergraduate students in applied AI techniques.
- Developed instructional material focused on practical model development and real-world applications.
- IEEE Phoenix Chapter Student Volunteer & Organizer

Honors & Awards

2025

■ **IEEE ICDM Scholarship**, IEEE International Conference on Data Mining, 2025.

2018

■ **Academic Scholarship**, Arizona State University, 2018.

2017–2018

■ **Best Undergraduate Researcher Award**, 2017–2018.